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Introduction to Prealgebra
Ch. 2 Exponents

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OUTLINE

- Squares
- Higher exponents
- Zero as an exponent
- Negative exponent
- Summary

SQUARES

- **Definition**
  - The **square** of a number *a*, written as a^2, is defined as

$$a^2 = a \cdot a.$$

  - A **perfect square** is the square of an integer.
    - E.g. 81 is the **perfect square** of 9.

SQUARES

- **The first 30 smallest perfect squares**

$$
\begin{array}{lllll}
0^2 = 0 & 1^2 = 1 & 2^2 = 4 & 3^2 = 9 & 4^2 = 16 \\
5^2 = 25 & 6^2 = 36 & 7^2 = 49 & 8^2 = 64 & 9^2 = 81 \\
10^2 = 100 & 11^2 = 121 & 12^2 = 144 & 13^2 = 169 & 14^2 = 196 \\
15^2 = 225 & 16^2 = 256 & 17^2 = 289 & 18^2 = 324 & 19^2 = 361 \\
20^2 = 400 & 21^2 = 441 & 22^2 = 484 & 23^2 = 529 & 24^2 = 576 \\
25^2 = 625 & 26^2 = 676 & 27^2 = 729 & 28^2 = 784 & 29^2 = 841
\end{array}
$$

SQUARES

- **Order of operations**

  - 1. Evaluate expression inside parenthesis first
  - 2. Compute powers
  - 3. Multiply and divide from left to right
  - 4. Add and subtract from left to right

SQUARES

- **Example**
  - What is the value of $(2x + 5)^2$ when $x = 3$?

SQUARES

- **Square of negation**

$$(-a)^2 = a^2.$$

- **Square are nonnegative**

Let a be a number. Then a^2 is nonnegative.

SQUARES

- **Example**
  - Given $x = -2$, find the value of $2x^2 + 3x + 4$.

SQUARES

- **Square of product**

$$(ab)^2 = a^2b^2.$$

  - Why?

SQUARES

- **Example**
  - Compute $8^2 \cdot 125^2$.

SQUARES

- **Example**
  - The sum $1^2 + 2^2 + 3^2 + 4^2 + \cdots + 25^2$ is 5525.
    Evaluate $2^2 + 4^2 + 6^2 + 8^2 + \cdots + 50^2$.

SQUARES

- **Square of reciprocal**

$$\left(\frac{1}{a}\right)^2 = \frac{1}{a^2}.$$

  - Why?

- **Example**
  - Compute $\left(\frac{1}{7}\right)^2$.

SQUARES

- **Square of quotient**

$$(a \div b)^2 = a^2 \div b^2.$$

  - Why?

- **Example**
  - Compute $369^2 \div 9^2$.

SQUARES

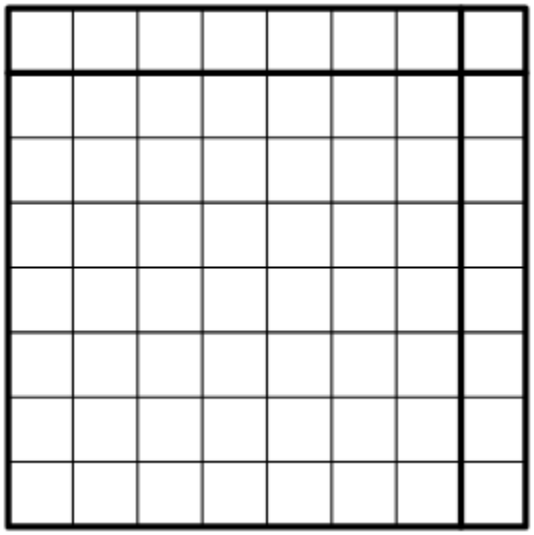
- **Quadratic form**

$$(a+b)^2 = a^2 + 2ab + b^2.$$

  - Why?

SQUARES

- **Example**
  - Use the picture below, explain why $8^2 = 7^2 + 2(7) + 1$.



SQUARES

- **Example**
  - Compute the following squares:

$$41^2 =$$

$$61^2 =$$

$$901^2 =$$

SQUARES

- **Example**
  - What perfect square is closest to 5,000?

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HIGHER EXPONENTS

- **Higher exponents examples**
  - 7 cubed: $7^3 = 7 \cdot 7 \cdot 7$.
  - 7 to the 4th: $7^4 = 7 \cdot 7 \cdot 7 \cdot 7$.
  - 7 to the 20th: $7^{20} = \underbrace{7 \times 7 \times \cdots \times 7}_{\text{20 copies of 7}}$.

HIGHER EXPONENTS

- **Definition**
  - Let *a* be any number and let *n* be a positive integer. The **power** a^n, pronounced as "*a* to the *n*", is defined as

$$a^n = \underbrace{a \times a \times \cdots \times a}_{n \text{ copies of } a}.$$

  - *a* is called the **base**.
  - *n* is called the **exponent**.

HIGHER EXPONENTS

- **Properties**
  - Let a and b be numbers. Let n be a positive integer.

$$(ab)^n = a^n b^n.$$

If b is nonzero, then $\left(\frac{1}{b}\right)^n = \frac{1}{b^n}$.

If b is nonzero, then $(a \div b)^n = a^n \div b^n$.

  - Why?

HIGHER EXPONENTS

- **Example**
  - a) Compute $(-4)^3$.
  - b) For how many integers n is n^3 between -50 and 50?

HIGHER EXPONENTS

- **Power of negation**
  - Let a be any number. Let n be a positive integer.

> If n is even, then $(-a)^n = a^n$.
> If n is odd, then $(-a)^n = -a^n$.

  - Why?

HIGHER EXPONENTS

- **Example**
  - Evaluate $(-1)^{(5^2)} + 1^{(2^5)}$.

HIGHER EXPONENTS

- **Product of powers (same base)**
  - Let a be any number. Let m and n be positive integers. Then

$$a^m a^n = a^{m+n}.$$

  - Review: product of powers (same exponent)

$$a^n b^n = (ab)^n.$$

HIGHER EXPONENTS

- **Example**
  - Express $5^{17} + 5^{17} + 5^{17} + 5^{17} + 5^{17}$ as a power of 5.

HIGHER EXPONENTS

- **Quotient of powers (same base)**

  - Let a be a nonzero number. Let m and n be positive integers. If m is greater than n, then

$$a^m \div a^n = a^{m-n}.$$

  - Review: quotient of powers (same exponent)

$$a^n \div b^n = (a \div b)^n.$$

HIGHER EXPONENTS

- **Example**
  - Calculate $4^{100} \div 4^{97}$.

HIGHER EXPONENTS

- **Example**
  - Evaluate the following two numbers.

$$(2^2)^3 =$$

$$2^{(2^3)} =$$

For a^{b^c}, we evaluate power from top to bottom, that is, $a^{b^c} = a^{(b^c)}$.

HIGHER EXPONENTS

- **Power of powers**
  - Let a be any number. Let m and n be positive integers. Then

$$(a^m)^n = a^{mn}.$$

  - Why?

HIGHER EXPONENTS

- **Example**
  - Express each of the following as a power of 2.

$$(2^7 \cdot 2^8) \div 2^3 =$$

$$(2^6)^4 \div 2^7 =$$

$$4^6 \div 8^2 =$$

HIGHER EXPONENTS

- **Example**
  - Compute $27^3 \div (9^2 \cdot 3^5)$.

HIGHER EXPONENTS

- **Example**
  - a) Express $11^{20,000}$ as a $10{,}000^{\text{th}}$ power.
  - b) Express $5^{30,000}$ as a $10{,}000^{\text{th}}$ power.
  - c) Express $2^{70,000}$ as a $10{,}000^{\text{th}}$ power.
  - d) Which of the numbers $11^{20,000}$, $5^{30,000}$, and $2^{70,000}$ is the greatest?

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ZERO AS AN EXPONENT

- **Example**
  - We know that if *m* and *n* are nonnegative integers and *m* is greater than *n*, then

$$3^{m-n} = 3^m \div 3^n.$$

  - Suppose this rule is true when *m* is equal to *n*. What is 3^0?

ZERO AS AN EXPONENT

- **Definition**
  - Let *a* be any number. Then a^0 is defined as

$$a^0 = 1.$$

. $a^{m-n} = a^m \div a^n.$

ZERO AS AN EXPONENT

- **Example**
  - Evaluate $6^0 + 6^1 + 6^2$.

ZERO AS AN EXPONENT

- **Example**
  - Let a be any number. Simplify $4a^0(4a)^0$.

ZERO AS AN EXPONENT

- **Example**

Let $P = (2-3-4+7)^{2347}$ and $Q = (-2+3+4-7)^{2347}$. What is the value of

$$(2+3+4+7)^{P+Q}?$$

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NEGATIVE EXPONENT

- **Example**
  - We know that if *m* and *n* are nonnegative integers and *m* is no less than *n*, then

$$3^{m-n} = 3^m \div 3^n.$$

  - Suppose this rule is true when *m* is less than *n*. What is 3^{-2}?

NEGATIVE EXPONENT

- **Definition**
  - Let *a* be any number. Let n be a positive integer. Then a^{-n} is defined as the reciprocal of a^n, so

$$a^{-n} = \frac{1}{a^n}.$$

NEGATIVE EXPONENT

- **Example**
  - Evaluate the following expressions.

$$1^{-5}.$$

$$10^{-4}.$$

$$2^{-3}.$$

$$56 \cdot 2^{-3}.$$

$$56 \div 2^{-3}.$$

NEGATIVE EXPONENTS

- **Product of powers (same base)**
  - Let a be any number. Let m and n be integers. Then

$$a^m a^n = a^{m+n}.$$

  - Review: product of powers (same exponent)

$$a^n b^n = (ab)^n.$$

NEGATIVE EXPONENT

- **Example**
  - Compute the following.

$$3^5 \cdot 3^{-5}.$$

$$3^6 \cdot 3^{-4}.$$

$$3^{15} \cdot 3^{-5} \cdot 3^{-4} \cdot 3^{-3}.$$

NEGATIVE EXPONENTS

- **Reciprocal of negative exponent**
  - Let a be nonzero and let n be a positive integer. Then

$$\frac{1}{a^{-n}} = a^n.$$

  - Why?

NEGATIVE EXPONENTS

- **Example**
  - Evaluate $\frac{1}{2^{-3}}$.
  - Evaluate $\frac{1}{5^{-2}}$.

NEGATIVE EXPONENTS

- **Power of reciprocal**
  - Let a be nonzero and let n be a positive integer. Then

$$\left(\frac{1}{a}\right)^{-n} = \frac{1}{a^{-n}} = a^n.$$

  - Why?

NEGATIVE EXPONENTS

- **Example**
  - Evaluate $\left(\frac{1}{2}\right)^{-1}$, $\left(\frac{1}{2}\right)^{-2}$, and $\left(\frac{1}{2}\right)^{-3}$.

NEGATIVE EXPONENTS

- **Example**
  - How are 2^{-3} and $\left(2^{-1}\right)^3$ related?

NEGATIVE EXPONENTS

- **Property**
  - Let a be nonzero and let n be a positive integer. Then

$$a^{-n} = \left(a^{-1}\right)^n.$$

  - Why?

NEGATIVE EXPONENTS

- **Example**
  - Let a and b be nonzero numbers. Explain why

$$a^{-6}b^{-6} = (ab)^{-6}.$$

NEGATIVE EXPONENTS

- **Properties**
  - Let a and b be numbers. Let m and n be integers.

$$a^m a^n = a^{m+n}.$$
$$a^n b^n = (ab)^n.$$
$$a^m \div a^n = a^{m-n}.$$
$$a^n \div b^n = (a \div b)^n.$$
$$(a^m)^n = a^{mn}.$$
$$\left(\frac{1}{a}\right)^n = \frac{1}{a^n}.$$
$$a^{-n} = \frac{1}{a^n}.$$

NEGATIVE EXPONENTS

- **Example**
  - Let x and y be nonzero numbers. Simplify

$$(x^4y^{-2})(x^{-1}y^5).$$

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SUMMARY

- **Definitions**

$$a^n = \underbrace{a \times a \times \cdots \times a}_{n \text{ copies of } a}.$$

$$a^{-n} = \frac{1}{a^n}.$$

SUMMARY

- **Properties**

$$a^m a^n = a^{m+n}.$$
$$a^n b^n = (ab)^n.$$
$$a^m \div a^n = a^{m-n}.$$
$$a^n \div b^n = (a \div b)^n.$$
$$(a^m)^n = a^{mn}.$$
$$\left(\frac{1}{a}\right)^n = \frac{1}{a^n}.$$
$$a^{-n} = \frac{1}{a^n}.$$